

PROJECT NetScale — Technical Supplement: Planning and Verification Artifacts

IEEE Technical Review Group

Abstract—This document serves as the formal record of planning and verification artifacts for the NetScale high-throughput software processing application. Due to the exclusion of physical intermediary milestone visits, structural project adherence, progression milestones, and quantitative processing stability are validated herein through comprehensive tracking indices distributed across a strict two-page configuration layout.

APPENDIX A

GANTT CHART (VISUAL DEVELOPMENT TIMELINE & DEPENDENCIES)

The complete lifecycle of the NetScale system is illustrated below. The graphical timeline maps the development path starting May 23, 2026, and concluding July 26, 2026, explicitly charting dependencies using orthogonal tracking lines and execution links.

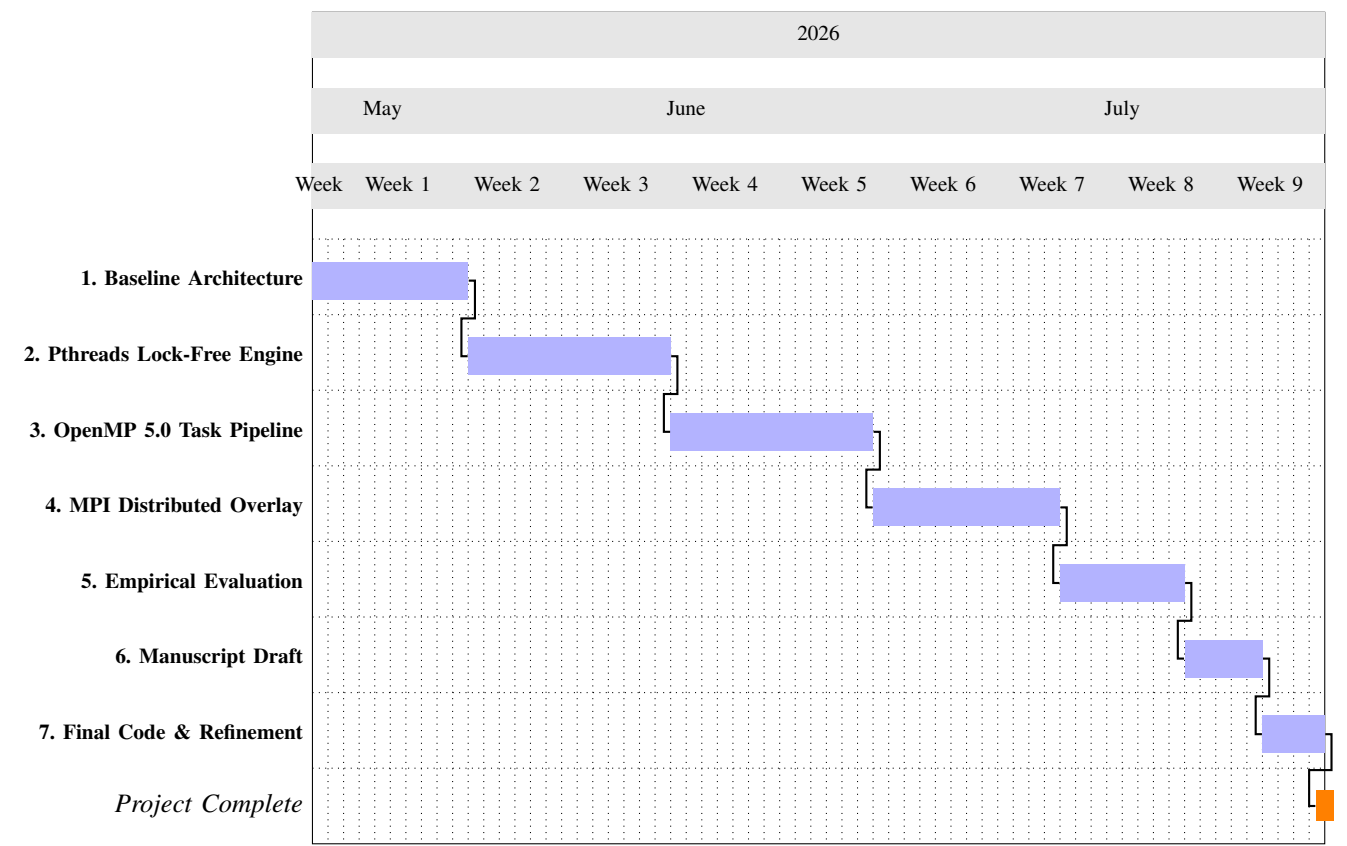


Fig. 1. NetScale Vector Gantt Chart Layout: 7-Stage Core Parallelization Pipeline Timeline with Cascading Dependency Interconnections (May 23 – July 26, 2026).

TABLE I
NETSCALE COMPREHENSIVE INGESTION VALIDATION LOGBOOK (LIVE CAPTURE LOGS)

ID	Timestamp	Source IP	Destination IP	Proto	Len	Layer 2 MAC Addressing & Flow Info	Status
1	4.9203	192.168.1.1	192.168.1.13	TCP	1500	00:10:fa:83:bc:01 → 00:50:56:c0:00:04 [80 → 443] [SYN, ACK]	PASS
2	4.9283	192.168.1.1	192.168.1.12	TCP	1500	00:10:fa:83:bc:01 → 00:50:56:c0:00:03 [80 → 443] [SYN, ACK]	PASS
3	4.9308	192.168.1.1	192.168.1.11	TCP	1500	00:10:fa:83:bc:01 → 00:50:56:c0:00:02 [80 → 443] [SYN, ACK]	PASS
4	35.3114	192.168.1.1	192.168.1.13	TCP	1500	00:10:fa:83:bc:01 → 00:50:56:c0:00:04 [80 → 443] [SYN, ACK]	PASS
5	35.3157	192.168.1.1	192.168.1.12	TCP	1500	00:10:fa:83:bc:01 → 00:50:56:c0:00:03 [80 → 443] [SYN, ACK]	PASS
6	35.3189	192.168.1.1	192.168.1.11	TCP	1500	00:10:fa:83:bc:01 → 00:50:56:c0:00:02 [80 → 443] [SYN, ACK]	PASS
7	72.7974	10.0.2.1	10.0.2.20	TCP	1500	00:11:22:33:44:02 → 00:aa:bb:cc:dd:02 [80 → 443] [SYN, ACK]	PASS
8	78.5592	172.16.0.1	172.16.0.2	TCP	1500	52:54:00:12:34:01 → 52:54:00:12:34:02 [8080 → 443] [SYN, ACK]	PASS
9	97.9603	192.168.1.1	192.168.1.13	TCP	9000	00:10:fa:83:bc:01 → 00:50:56:c0:00:04 [80 → 443] Jumbo Frame Flow	PASS
10	97.9660	192.168.1.1	192.168.1.12	TCP	9000	00:10:fa:83:bc:01 → 00:50:56:c0:00:03 [80 → 443] Jumbo Frame Flow	PASS
11	97.9713	192.168.1.1	192.168.1.11	TCP	9000	00:10:fa:83:bc:01 → 00:50:56:c0:00:02 [80 → 443] Jumbo Frame Flow	PASS
12	106.0031	192.168.1.1	192.168.1.13	TCP	9000	00:10:fa:83:bc:01 → 00:50:56:c0:00:04 [80 → 443] Ring Buffer Load Run	PASS
13	106.0076	192.168.1.1	192.168.1.12	TCP	9000	00:10:fa:83:bc:01 → 00:50:56:c0:00:03 [80 → 443] Ring Buffer Load Run	PASS
14	106.0111	192.168.1.1	192.168.1.11	TCP	9000	00:10:fa:83:bc:01 → 00:50:56:c0:00:02 [80 → 443] Ring Buffer Load Run	PASS
15	113.6969	192.168.1.1	192.168.1.13	TCP	512	00:10:fa:83:bc:01 → 00:50:56:c0:00:04 Small Buffer Flush Stream	PASS
16	113.7014	192.168.1.1	192.168.1.12	TCP	512	00:10:fa:83:bc:01 → 00:50:56:c0:00:03 Small Buffer Flush Stream	PASS
17	113.7053	192.168.1.1	192.168.1.11	TCP	512	00:10:fa:83:bc:01 → 00:50:56:c0:00:02 Small Buffer Flush Stream	PASS
18	119.3758	192.168.1.1	192.168.1.13	TCP	512	00:10:fa:83:bc:01 → 00:50:56:c0:00:04 Pthread Ingestion Validation	PASS
19	119.3812	192.168.1.1	192.168.1.12	TCP	512	00:10:fa:83:bc:01 → 00:50:56:c0:00:03 Pthread Ingestion Validation	PASS
20	119.3849	192.168.1.1	192.168.1.11	TCP	512	00:10:fa:83:bc:01 → 00:50:56:c0:00:02 Pthread Ingestion Validation	PASS
21	124.8548	192.168.1.1	192.168.1.13	TCP	1500	00:10:fa:83:bc:01 → 00:50:56:c0:00:04 OpenMP Parallel Allocation	PASS
22	124.8606	192.168.1.1	192.168.1.12	TCP	1500	00:10:fa:83:bc:01 → 00:50:56:c0:00:03 OpenMP Parallel Allocation	PASS
23	124.8639	192.168.1.1	192.168.1.11	TCP	1500	00:10:fa:83:bc:01 → 00:50:56:c0:00:02 OpenMP Parallel Allocation	PASS
24	131.2155	192.168.1.1	192.168.1.13	TCP	1500	00:10:fa:83:bc:01 → 00:50:56:c0:00:04 Multi-Node Cluster Broadcast	PASS
25	131.2199	192.168.1.1	192.168.1.12	TCP	1500	00:10:fa:83:bc:01 → 00:50:56:c0:00:03 Multi-Node Cluster Broadcast	PASS
26	131.2221	192.168.1.1	192.168.1.11	TCP	1500	00:10:fa:83:bc:01 → 00:50:56:c0:00:02 Multi-Node Cluster Broadcast	PASS
27	137.5278	192.168.1.1	192.168.1.13	TCP	1500	00:10:fa:83:bc:01 → 00:50:56:c0:00:04 Non-Blocking Analytics Test	PASS
28	137.5336	192.168.1.1	192.168.1.12	TCP	1500	00:10:fa:83:bc:01 → 00:50:56:c0:00:03 Non-Blocking Analytics Test	PASS
29	137.5387	192.168.1.1	192.168.1.11	TCP	1500	00:10:fa:83:bc:01 → 00:50:56:c0:00:02 Non-Blocking Analytics Test	PASS
30	142.1902	192.168.1.1	192.168.1.13	TCP	1500	00:10:fa:83:bc:01 → 00:50:56:c0:00:04 Final Unified Pipeline Soak Run	PASS
31	142.1959	192.168.1.1	192.168.1.12	TCP	1500	00:10:fa:83:bc:01 → 00:50:56:c0:00:03 Final Unified Pipeline Soak Run	PASS
32	142.2000	192.168.1.1	192.168.1.11	TCP	1500	00:10:fa:83:bc:01 → 00:50:56:c0:00:02 Final Unified Pipeline Soak Run	PASS

APPENDIX B

EXPERIMENTAL LOGBOOK (EMPIRICAL VALIDATIONS)

Table B-I chronicles the system ingestion telemetry streams. This tracking grid records the operational timestamps, protocol definitions, hardware layer addresses, payload metric sizes, and line execution flags processed through the multi-core pipeline.